

SDG 3 Indicator Text Classification

A Multi-Label NLP Classification System

Group 8 | Assignment 2

Kayonga Elvis · Nformi Modestine Girbong · [Person C] · [Person D]

Machine learning techniques 1

African leadership university

01/06/2026

Abstract

[150–250 word summary: problem statement, dataset, methods, key findings (best Hamming Loss), and main conclusions. Write this last.]

1. Introduction

1.1 Problem Statement

[Introduce SDG 3 (Good Health and Well-being) and its indicator framework. Explain why automated text classification of development documents is valuable. State the multi-label classification problem clearly: given a text sample, predict the subset of SDG 3 indicators relevant to it.]

1.2 Objectives

- Build a reproducible multi-label classification pipeline for SDG 3 indicators.
- Systematically compare preprocessing strategies, feature representations, and model architectures.
- Minimize Hamming Loss on a held-out test set.
- Document insights and limitations for future practitioners.

1.3 Report Structure

1. Introduction

1.1 Problem Statement

1.2 Objectives

1.3 Report Structure

2. Literature Review

2.1 Multi-Label Text Classification

2.2 Text Representation Techniques

2.3 Evaluation for Multi-Label Classification

2.4 SDG Text Classification Prior Work

3. Dataset and Exploratory Data Analysis

3.1 Dataset Description

3.2 Label Distribution Analysis

3.3 Text Characteristics

3.4 Co-occurrence and Label Correlations

4. Methodology

4.1 Preprocessing Pipeline

4.2 Feature Engineering Traditional Methods

4.3 Multi-Label Classification Framework

4.4 Baseline Models

4.5 Advanced Models and Embeddings

4.6 Evaluation Methodology

5. Experimentation

5.1 Experiment Log

5.2 Experiment Details

6. Results

6.1 Model Comparison Table

6.2 Learning Curves

6.3 Per-Label Performance

6.4 Confusion Matrix / Classification Visualization

6.5 Final Test Predictions

7. Discussion

7.1 Why Certain Approaches Performed Better

7.2 Tradeoffs Observed

7.3 Preprocessing and Feature Engineering Impact

7.4 Dataset and Methodology Limitations

7.5 Lessons Learned

8. Ethical Considerations and Responsible AI

9. Conclusion and Future Work

9.1 Summary

9.2 Future Work

Reference

appendix

2. Literature Review

2.1 Multi-Label Text Classification

[Discuss foundational and recent work on multi-label classification. Cite key papers on problem transformation strategies (Binary Relevance, Classifier Chains, Label Powerset) and algorithm adaptation methods. Example: Tsoumakas & Katakis (2007) survey of multi-label learning.]

2.2 Text Representation Techniques

[Review TF-IDF, Bag-of-Words, word embeddings (Word2Vec, GloVe), and transformer-based representations (BERT, RoBERTa). Discuss trade-offs between sparse

and dense representations for short/long documents. [Person B focus: justify why you chose TF-IDF for baselines.]]

2.3 Evaluation for Multi-Label Classification

[Review Hamming Loss, Micro/Macro F1, Subset Accuracy, and Jaccard Similarity. Explain why Hamming Loss is particularly appropriate here. Cite Sorower (2010) or similar surveys.]

2.4 SDG Text Classification Prior Work

[Cite any existing systems for SDG text classification or sustainable development NLP. Useful sources: Transformers for SDGs (Müller-Eberstein et al., 2022), OSDG tool, or similar.]

3. Dataset and Exploratory Data Analysis

Note: Sections 3.1–3.3 are primarily Person A's contribution. Person B contributes to Section 3.4 regarding preprocessing choices that inform feature engineering.

3.1 Dataset Description

[Describe Devex_train.csv and Devex_test_questions.csv. Report: total samples (~3,000), number of SDG 3 indicator labels, text source types (tenders, reports, news articles, etc.), average/median document length (tokens/characters).]

3.2 Label Distribution Analysis

[[INSERT Figure: Bar chart of label frequencies across all SDG 3 indicators — Person A] Discuss class imbalance: which indicators are over/under-represented? How does this inform the choice of evaluation metrics and potential balancing strategies?]

3.3 Text Characteristics

[[INSERT Figures: Document length distribution histogram; word cloud of most frequent terms — Person A] Discuss vocabulary size, average sentence count, and any domain-specific language patterns observed.]

3.4 Co-occurrence and Label Correlations

[[INSERT Figure: Label co-occurrence heatmap — Person A] Which indicators frequently appear together? Does this suggest that Classifier Chains might outperform Binary Relevance?]

4. Methodology

4.1 Preprocessing Pipeline

Note: Person A leads 4.1; Person B leads 4.2 and 4.3.

[Describe the complete preprocessing pipeline: • Lowercasing • Punctuation/special character removal • Tokenization approach (whitespace, NLTK, SpaCy) • Stop word removal (which list? domain-specific additions?) • Stemming vs. lemmatization (and justification for the choice) • Handling of numerical tokens • Treatment of domain-specific acronyms (e.g., 'SDG', 'HIV', 'WHO')]

4.2 Feature Engineering — Traditional Methods

★ Person B — Core Responsibility

[Explain TF-IDF vectorization: mathematical formulation, vocabulary size decisions, min_df / max_df cutoffs, sublinear_tf usage, and n-gram range experimentation. Discuss Bag-of-Words as a baseline comparison. Justify hyperparameter choices based on EDA findings (e.g., vocabulary size relative to dataset size).]

4.3 Multi-Label Classification Framework

★ Person B — Core Responsibility

[Describe the Binary Relevance (One-vs-Rest) strategy used as the primary multi-label framework. Briefly discuss alternatives (Classifier Chains, Label Powerset) and justify the choice. Explain the thresholding strategy for converting probability scores to binary labels (default 0.5 vs. optimized threshold).]

4.4 Baseline Models

★ Person B — Core Responsibility

[Describe each baseline model: • Logistic Regression (solver, C, max_iter) • Linear SVM (C, kernel) • Random Forest (n_estimators, max_depth) For each: mathematical intuition, key hyperparameters, and justification for inclusion.]

4.5 Advanced Models and Embeddings

Note: Person C leads this section.

[[Person C: Describe Word2Vec / GloVe / BERT embedding approach, neural network architecture, training setup.]]

4.6 Evaluation Methodology

★ Person B — Core Responsibility

[Define each evaluation metric with formula and interpretation: • Hamming Loss: fraction of incorrectly predicted labels — the primary assignment metric. Formula: $HL = (1/N \cdot L) \sum$

$XOR(y_i, \hat{y}_i)$ • Micro F1: aggregates TP/FP/FN across all labels — useful when overall performance matters. • Macro F1: unweighted mean per-label F1 — highlights performance on rare indicators. • Subset Accuracy: exact match ratio — strict metric for complete prediction correctness. Describe train/validation split strategy (e.g., 80/20 stratified). Discuss why stratified splitting is important given label imbalance.]

5. Experimentation

This section documents all experiments conducted, following a systematic progression where each experiment informs the next. The primary evaluation metric is Hamming Loss (lower = better).

5.1 Experiment Log

[[Fill in the table below. Aim for 6–10 meaningful experiments. Rows are pre-populated for Person B's expected baseline experiments; add/extend for Person C's advanced experiments.]]

#	Change Made	Rationale	Outcome (Hamming Loss / F1)	Insight
Exp 1				
Exp 2				
Exp 3				
Exp 4				
Exp 5				
Exp 6				
Exp 7				
Exp 8				

5.2 Experiment Details

Experiment 1: Baseline — Logistic Regression with TF-IDF (Unigrams)

[What changed: Initial baseline. No prior experiment. Why: Establish a lower bound. TF-IDF + LR is a well-known strong baseline for text classification (Joulin et al., 2017). Setup: TF-IDF (max_features=10,000, unigrams), OneVsRestClassifier(LogisticRegression(C=1.0,

max_iter=1000)), threshold=0.5. Outcome: Hamming Loss = [X.XXX], Micro F1 = [X.XX], Macro F1 = [X.XX] Insight: [What did you observe? Which labels were hardest? Was there a precision/recall tradeoff?]

Experiment 2: Effect of N-gram Range (Unigrams + Bigrams)

[What changed: Added bigrams (ngram_range=(1,2)) to TF-IDF. Why: Exp 1 showed the model missed multi-word domain phrases (e.g., 'maternal mortality', 'universal health coverage'). Bigrams may capture these. Outcome: [Results] Insight: [Did bigrams help? Increased vocabulary size tradeoff?]

Experiment 3: Vocabulary Size Tuning

[What changed: Varied max_features {5000, 10000, 20000, 50000}. Why: Exp 2 suggested vocabulary coverage matters; testing the optimal size. Outcome: [Results table or best value] Insight: [Diminishing returns? Overfitting with very large vocabulary?]

Experiment 4: Classifier Comparison (SVM vs. LR vs. RF)

[What changed: Replaced Logistic Regression with LinearSVC and RandomForest using best TF-IDF config from Exp 3. Why: Explore whether a different decision boundary (SVM margin) or ensemble approach (RF) improves performance. Outcome: [Comparative results] Insight: [Why did one outperform the others? Consider data dimensionality and linear separability.]]

Experiment 5: Threshold Optimization

[What changed: Replaced default threshold (0.5) with a label-optimized threshold using validation F1. Why: Multi-label classification with imbalanced classes often benefits from per-label thresholds (Yang, 2001). Outcome: [Results] Insight: [Which labels had optimal thresholds far from 0.5? What does this tell us about class imbalance?]]

Experiment 6: Addressing Class Imbalance

[What changed: Applied class_weight='balanced' in Logistic Regression and/or used oversampling (MLSMOTE or RandomOverSampler). Why: EDA (Section 3.2) revealed significant label imbalance. Exp 4 showed poor Macro F1, suggesting rare labels are under-predicted. Outcome: [Results] Insight: [Did balancing improve Macro F1 at the cost of Hamming Loss? Discuss the tradeoff.]]

Experiments 7–10: Advanced Models

Note: Person C completes experiments 7–10 following the same documentation format above.

[[Person C: Document advanced embedding experiments here — Word2Vec/GloVe aggregation, BERT fine-tuning, neural architectures, hyperparameter search.]]

6. Results

6.1 Model Comparison Table

Model / Config	Hamming Loss ↓	Micro F1 ↑	Macro F1 ↑	Subset Accuracy ↑	Notes
LR + TF-IDF (Baseline)					Person B
SVM + TF-IDF					Person B
Random Forest + TF-IDF					Person B
LR + TF-IDF + n-grams					Person B
LR + Oversampling					Person B
Advanced Model (Emb.)					Person C
Best Configuration					

6.2 Learning Curves

[[INSERT Figure: Training vs. validation Hamming Loss over training iterations for best model(s) — Person B/C] Note any signs of overfitting or underfitting. Learning curves validate that the reported results are not artifacts of lucky train/val splits.]

6.3 Per-Label Performance

[[INSERT Figure: Per-SDG-indicator F1 bar chart for best model — Person D] Highlight which indicators are classified well vs. poorly. Connect to label frequency distribution from Section 3.2.]

6.4 Confusion Matrix / Classification Visualization

[[INSERT Figure: Heatmap of per-label prediction accuracy or confusion matrix for each indicator — Person D] For multi-label problems, a per-class precision-recall curve or classification report table is often more informative than a traditional confusion matrix.]

6.5 Final Test Predictions

[Describe how predictions were generated for Devex_test_questions.csv using the best model configuration. Report the number of test samples and any post-processing applied to predictions (e.g., ensuring at least one label per sample).]

7. Discussion

Note: Person D leads this section. All members review for accuracy.

7.1 Why Certain Approaches Performed Better

[Interpret the experiment results. Why did [best model] outperform [baseline]? Reference specific properties: e.g., linear separability of the feature space, the model's ability to handle high-dimensional sparse vectors, or the semantic richness of embeddings.]

7.2 Tradeoffs Observed

[Discuss key tradeoffs: • Hamming Loss vs. Macro F1 (balancing strategies often improve rare-label recall but hurt overall Hamming Loss) • Vocabulary size vs. computational cost • Model complexity vs. interpretability • Threshold tuning vs. robustness]

7.3 Preprocessing and Feature Engineering Impact

[Quantify the impact of preprocessing decisions. Did removing/keeping domain acronyms matter? Did stemming vs. lemmatization affect results measurably? Connect specific experiments to these findings.]

7.4 Dataset and Methodology Limitations

[Honest assessment of limitations: • Label quality: were ground-truth labels perfect or subjective? • Dataset size (~3,000 samples): sufficient for BERT fine-tuning? • Domain shift: training on development docs but some test docs may be different domains • Multi-label evaluation: Hamming Loss alone may not capture all aspects of model quality]

7.5 Lessons Learned

[What would you do differently if starting over? What surprised you? What did not work as expected and why?]

8. Ethical Considerations and Responsible AI

[Discuss: • Bias in training data: are certain geographies, languages, or document types over-represented? How might this affect predictions for underrepresented communities? • Consequences of misclassification: if a development program is wrongly tagged, could it affect resource allocation? • Model transparency: are the decisions interpretable by domain experts (public health practitioners)? • Data provenance: was the training data collected

ethically? Are there privacy concerns? • Fairness: does the model perform equally well across all SDG 3 indicators, or systematically worse for some?]

9. Conclusion and Future Work

9.1 Summary

[Summarize the project: problem, approach, best result (Hamming Loss = X.XXX with [model]), and key insights in 2–3 paragraphs.]

9.2 Future Work

- Fine-tune domain-specific transformers (e.g., PubMedBERT, DevelopmentBERT) on larger SDG corpora.
- Explore Classifier Chains to model label dependencies observed in the co-occurrence analysis.
- Incorporate document-level features (source type, year, geographic focus) as auxiliary inputs.
- Active learning to efficiently label additional ambiguous samples.
- Ensemble methods combining TF-IDF models with embedding-based models.

References

[[Format all references in IEEE or APA style — Person D] Suggested starting references: • Tsoumakas, G., & Katakis, I. (2007). Multi-label classification: An overview. • Müller-Eberstein, M., et al. (2022). Towards a framework for predicting SDGs with transformers. • Sorower, M. S. (2010). A literature survey on algorithms for multi-label learning. • Yang, Y. (2001). A study of thresholding strategies for text categorization. • Joulin, A., et al. (2017). Bag of tricks for efficient text classification. (FastText) • Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers.]

Appendices

Appendix A: Member Contributions

Member	Contributions	Role	% Effort
--------	---------------	------	----------

Person A (Name)	EDA, preprocessing pipeline, visualizations	Data Architect	25%
Person B (Your Name)	TF-IDF, baseline models, evaluation metrics section	Feature Engineer & Baseline Modeler	25%
Person C (Name)	Advanced embeddings, deep learning models	Advanced Modeler	25%
Person D (Name)	Results analysis, GitHub repo, report integration	Analyst & Repo Manager	25%

Link to group contribution tracker:

[<https://docs.google.com/spreadsheets/...>]

Appendix B: GitHub Repository

[Repository URL: [https://github.com/\[username\]/\[repo-name\]](https://github.com/[username]/[repo-name]) Demo Video URL: [YouTube or Drive link]]

Appendix C: Supplementary Figures

[[Insert any additional EDA plots, per-label learning curves, or ablation study tables that support the main text but would disrupt flow if placed inline.]]

SDG 3 Indicator Text Classification

A Multi-Label NLP Classification System

Group [Name] | Assignment 2

[Person A] · [Person B / Your Name] · [Person C] · [Person D]

[Course Name & Code] | [University] | [Date]

Abstract

[150–250 word summary: problem statement, dataset, methods, key findings (best Hamming Loss), and main conclusions. Write this last.]

1. Introduction

1.1 Problem Statement

[Introduce SDG 3 (Good Health and Well-being) and its indicator framework. Explain why automated text classification of development documents is valuable. State the multi-label classification problem clearly: given a text sample, predict the subset of SDG 3 indicators relevant to it.]

1.2 Objectives

- Build a reproducible multi-label classification pipeline for SDG 3 indicators.
- Systematically compare preprocessing strategies, feature representations, and model architectures.
- Minimize Hamming Loss on a held-out test set.
- Document insights and limitations for future practitioners.

1.3 Report Structure

[Briefly describe what each subsequent section covers.]

2. Literature Review

2.1 Multi-Label Text Classification

[Discuss foundational and recent work on multi-label classification. Cite key papers on problem transformation strategies (Binary Relevance, Classifier Chains, Label Powerset) and algorithm adaptation methods. Example: Tsoumakas & Katakis (2007) survey of multi-label learning.]

2.2 Text Representation Techniques

[Review TF-IDF, Bag-of-Words, word embeddings (Word2Vec, GloVe), and transformer-based representations (BERT, RoBERTa). Discuss trade-offs between sparse

and dense representations for short/long documents. [Person B focus: justify why you chose TF-IDF for baselines.]]

2.3 Evaluation for Multi-Label Classification

[Review Hamming Loss, Micro/Macro F1, Subset Accuracy, and Jaccard Similarity. Explain why Hamming Loss is particularly appropriate here. Cite Sorower (2010) or similar surveys.]

2.4 SDG Text Classification Prior Work

[Cite any existing systems for SDG text classification or sustainable development NLP. Useful sources: Transformers for SDGs (Müller-Eberstein et al., 2022), OSDG tool, or similar.]

3. Dataset and Exploratory Data Analysis

Note: Sections 3.1–3.3 are primarily Person A's contribution. Person B contributes to Section 3.4 regarding preprocessing choices that inform feature engineering.

3.1 Dataset Description

[Describe Devex_train.csv and Devex_test_questions.csv. Report: total samples (~3,000), number of SDG 3 indicator labels, text source types (tenders, reports, news articles, etc.), average/median document length (tokens/characters).]

3.2 Label Distribution Analysis

[[INSERT Figure: Bar chart of label frequencies across all SDG 3 indicators — Person A] Discuss class imbalance: which indicators are over/under-represented? How does this inform the choice of evaluation metrics and potential balancing strategies?]

3.3 Text Characteristics

[[INSERT Figures: Document length distribution histogram; word cloud of most frequent terms — Person A] Discuss vocabulary size, average sentence count, and any domain-specific language patterns observed.]

3.4 Co-occurrence and Label Correlations

[[INSERT Figure: Label co-occurrence heatmap — Person A] Which indicators frequently appear together? Does this suggest that Classifier Chains might outperform Binary Relevance?]

4. Methodology

4.1 Preprocessing Pipeline

Note: Person A leads 4.1; Person B leads 4.2 and 4.3.

[Describe the complete preprocessing pipeline: • Lowercasing • Punctuation/special character removal • Tokenization approach (whitespace, NLTK, SpaCy) • Stop word removal (which list? domain-specific additions?) • Stemming vs. lemmatization (and justification for the choice) • Handling of numerical tokens • Treatment of domain-specific acronyms (e.g., 'SDG', 'HIV', 'WHO')]

4.2 Feature Engineering — Traditional Methods

★ Person B — Core Responsibility

[Explain TF-IDF vectorization: mathematical formulation, vocabulary size decisions, min_df / max_df cutoffs, sublinear_tf usage, and n-gram range experimentation. Discuss Bag-of-Words as a baseline comparison. Justify hyperparameter choices based on EDA findings (e.g., vocabulary size relative to dataset size).]

4.3 Multi-Label Classification Framework

★ Person B — Core Responsibility

[Describe the Binary Relevance (One-vs-Rest) strategy used as the primary multi-label framework. Briefly discuss alternatives (Classifier Chains, Label Powerset) and justify the choice. Explain the thresholding strategy for converting probability scores to binary labels (default 0.5 vs. optimized threshold).]

4.4 Baseline Models

★ Person B — Core Responsibility

[Describe each baseline model: • Logistic Regression (solver, C, max_iter) • Linear SVM (C, kernel) • Random Forest (n_estimators, max_depth) For each: mathematical intuition, key hyperparameters, and justification for inclusion.]

4.5 Advanced Models and Embeddings

Note: Person C leads this section.

[[Person C: Describe Word2Vec / GloVe / BERT embedding approach, neural network architecture, training setup.]]

4.6 Evaluation Methodology

★ Person B — Core Responsibility

[Define each evaluation metric with formula and interpretation: • Hamming Loss: fraction of incorrectly predicted labels — the primary assignment metric. Formula: $HL = (1/N \cdot L) \sum$

$XOR(y_i, \hat{y}_i)$ • Micro F1: aggregates TP/FP/FN across all labels — useful when overall performance matters. • Macro F1: unweighted mean per-label F1 — highlights performance on rare indicators. • Subset Accuracy: exact match ratio — strict metric for complete prediction correctness. Describe train/validation split strategy (e.g., 80/20 stratified). Discuss why stratified splitting is important given label imbalance.]

5. Experimentation

This section documents all experiments conducted, following a systematic progression where each experiment informs the next. The primary evaluation metric is Hamming Loss (lower = better).

5.1 Experiment Log

[[Fill in the table below. Aim for 6–10 meaningful experiments. Rows are pre-populated for Person B's expected baseline experiments; add/extend for Person C's advanced experiments.]]

#	Change Made	Rationale	Outcome (Hamming Loss / F1)	Insight
Exp 1				
Exp 2				
Exp 3				
Exp 4				
Exp 5				
Exp 6				
Exp 7				
Exp 8				

5.2 Experiment Details

Experiment 1: Baseline — Logistic Regression with TF-IDF (Unigrams)

[What changed: Initial baseline. No prior experiment. Why: Establish a lower bound. TF-IDF + LR is a well-known strong baseline for text classification (Joulin et al., 2017). Setup: TF-IDF (max_features=10,000, unigrams), OneVsRestClassifier(LogisticRegression(C=1.0,

max_iter=1000)), threshold=0.5. Outcome: Hamming Loss = [X.XXX], Micro F1 = [X.XX], Macro F1 = [X.XX] Insight: [What did you observe? Which labels were hardest? Was there a precision/recall tradeoff?]

Experiment 2: Effect of N-gram Range (Unigrams + Bigrams)

[What changed: Added bigrams (ngram_range=(1,2)) to TF-IDF. Why: Exp 1 showed the model missed multi-word domain phrases (e.g., 'maternal mortality', 'universal health coverage'). Bigrams may capture these. Outcome: [Results] Insight: [Did bigrams help? Increased vocabulary size tradeoff?]

Experiment 3: Vocabulary Size Tuning

[What changed: Varied max_features {5000, 10000, 20000, 50000}. Why: Exp 2 suggested vocabulary coverage matters; testing the optimal size. Outcome: [Results table or best value] Insight: [Diminishing returns? Overfitting with very large vocabulary?]

Experiment 4: Classifier Comparison (SVM vs. LR vs. RF)

[What changed: Replaced Logistic Regression with LinearSVC and RandomForest using best TF-IDF config from Exp 3. Why: Explore whether a different decision boundary (SVM margin) or ensemble approach (RF) improves performance. Outcome: [Comparative results] Insight: [Why did one outperform the others? Consider data dimensionality and linear separability.]]

Experiment 5: Threshold Optimization

[What changed: Replaced default threshold (0.5) with a label-optimized threshold using validation F1. Why: Multi-label classification with imbalanced classes often benefits from per-label thresholds (Yang, 2001). Outcome: [Results] Insight: [Which labels had optimal thresholds far from 0.5? What does this tell us about class imbalance?]]

Experiment 6: Addressing Class Imbalance

[What changed: Applied class_weight='balanced' in Logistic Regression and/or used oversampling (MLSMOTE or RandomOverSampler). Why: EDA (Section 3.2) revealed significant label imbalance. Exp 4 showed poor Macro F1, suggesting rare labels are under-predicted. Outcome: [Results] Insight: [Did balancing improve Macro F1 at the cost of Hamming Loss? Discuss the tradeoff.]]

Experiments 7–10: Advanced Models

Note: Person C completes experiments 7–10 following the same documentation format above.

[[Person C: Document advanced embedding experiments here — Word2Vec/GloVe aggregation, BERT fine-tuning, neural architectures, hyperparameter search.]]

6. Results

6.1 Model Comparison Table

Model / Config	Hamming Loss ↓	Micro F1 ↑	Macro F1 ↑	Subset Accuracy ↑	Notes
LR + TF-IDF (Baseline)					Person B
SVM + TF-IDF					Person B
Random Forest + TF-IDF					Person B
LR + TF-IDF + n-grams					Person B
LR + Oversampling					Person B
Advanced Model (Emb.)					Person C
Best Configuration					All

6.2 Learning Curves

[[INSERT Figure: Training vs. validation Hamming Loss over training iterations for best model(s) — Person B/C] Note any signs of overfitting or underfitting. Learning curves validate that the reported results are not artifacts of lucky train/val splits.]

6.3 Per-Label Performance

[[INSERT Figure: Per-SDG-indicator F1 bar chart for best model — Person D] Highlight which indicators are classified well vs. poorly. Connect to label frequency distribution from Section 3.2.]

6.4 Confusion Matrix / Classification Visualization

[[INSERT Figure: Heatmap of per-label prediction accuracy or confusion matrix for each indicator — Person D] For multi-label problems, a per-class precision-recall curve or classification report table is often more informative than a traditional confusion matrix.]

6.5 Final Test Predictions

[Describe how predictions were generated for Devex_test_questions.csv using the best model configuration. Report the number of test samples and any post-processing applied to predictions (e.g., ensuring at least one label per sample).]

7. Discussion

Note: Person D leads this section. All members review for accuracy.

7.1 Why Certain Approaches Performed Better

[Interpret the experiment results. Why did [best model] outperform [baseline]? Reference specific properties: e.g., linear separability of the feature space, the model's ability to handle high-dimensional sparse vectors, or the semantic richness of embeddings.]

7.2 Tradeoffs Observed

[Discuss key tradeoffs: • Hamming Loss vs. Macro F1 (balancing strategies often improve rare-label recall but hurt overall Hamming Loss) • Vocabulary size vs. computational cost • Model complexity vs. interpretability • Threshold tuning vs. robustness]

7.3 Preprocessing and Feature Engineering Impact

[Quantify the impact of preprocessing decisions. Did removing/keeping domain acronyms matter? Did stemming vs. lemmatization affect results measurably? Connect specific experiments to these findings.]

7.4 Dataset and Methodology Limitations

[Honest assessment of limitations: • Label quality: were ground-truth labels perfect or subjective? • Dataset size (~3,000 samples): sufficient for BERT fine-tuning? • Domain shift: training on development docs but some test docs may be different domains • Multi-label evaluation: Hamming Loss alone may not capture all aspects of model quality]

7.5 Lessons Learned

[What would you do differently if starting over? What surprised you? What did not work as expected and why?]

8. Ethical Considerations and Responsible AI

[Discuss: • Bias in training data: are certain geographies, languages, or document types over-represented? How might this affect predictions for underrepresented communities? • Consequences of misclassification: if a development program is wrongly tagged, could it affect resource allocation? • Model transparency: are the decisions interpretable by domain experts (public health practitioners)? • Data provenance: was the training data collected

ethically? Are there privacy concerns? • Fairness: does the model perform equally well across all SDG 3 indicators, or systematically worse for some?]

9. Conclusion and Future Work

9.1 Summary

[Summarize the project: problem, approach, best result (Hamming Loss = X.XXX with [model]), and key insights in 2–3 paragraphs.]

9.2 Future Work

- Fine-tune domain-specific transformers (e.g., PubMedBERT, DevelopmentBERT) on larger SDG corpora.
- Explore Classifier Chains to model label dependencies observed in the co-occurrence analysis.
- Incorporate document-level features (source type, year, geographic focus) as auxiliary inputs.
- Active learning to efficiently label additional ambiguous samples.
- Ensemble methods combining TF-IDF models with embedding-based models.

References

[[Format all references in IEEE or APA style — Person D] Suggested starting references: • Tsoumakas, G., & Katakis, I. (2007). Multi-label classification: An overview. • Müller-Eberstein, M., et al. (2022). Towards a framework for predicting SDGs with transformers. • Sorower, M. S. (2010). A literature survey on algorithms for multi-label learning. • Yang, Y. (2001). A study of thresholding strategies for text categorization. • Joulin, A., et al. (2017). Bag of tricks for efficient text classification. (FastText) • Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers.]

Appendices

Appendix A: Member Contributions

Member	Contributions	Role	% Effort
--------	---------------	------	----------

Person A (Name)	EDA, preprocessing pipeline, visualizations	Data Architect	25%
Nformi Modestine Girbong	TF-IDF, baseline models, evaluation metrics section	Feature Engineer & Baseline Modeler	25%
Person C (Name)	Advanced embeddings, deep learning models	Advanced Modeler	25%
Person D (Name)	Results analysis, GitHub repo, report integration	Analyst & Repo Manager	25%

Link to group contribution tracker:

<https://docs.google.com/spreadsheets/d/1dbr1B950tA2QlfaoDuuHlwsv42XjWem5CnBcfrX4F58/edit?gid=0#gid=0>

Appendix B: GitHub Repository

<https://github.com/NFORMII/SDG-Indicator-Text-Classification>

Appendix C: Supplementary Figures

[[Insert any additional EDA plots, per-label learning curves, or ablation study tables that support the main text but would disrupt flow if placed inline.]]